

- ✓ The assignment must be submitted within the deadline. **Late submissions will be penalized.**
- ✓ Submission of softcopy of the assignment is mandatory.
- ✓ Submit your assignment through this link - <https://forms.gle/7ASShRziaw8nkWv98>

■ **Question 1 of 4** [14 marks]

34	22	12	10	9
14	13	$k + 1$	11	13
17	$k - 1$	k	$k - 1$	3
28	29	$k + 1$	21	33
30	27	13	25	31

Grayscale Image, I

1	0	-1
2	0	-2
1	0	-1

Kernel 1, k_1

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Kernel 2, k_2

Use $k = \text{mod}(\text{Student ID}, 245) + 1$ for this question.

For all operations use the truncated variants. You have to work out only one operation for all the questions.

- (a) [3 marks] Find $I * k_1$
- (b) [3 marks] Find $I * k_2$
- (c) [3 marks] Find $I \star k_1$
- (d) [3 marks] Find $I \star k_2$
- (e) [2 marks] Check if $I * k_1 == I \star k_1$ and $I * k_2 == I \star k_2$? Provide explanation against your answer.

■ Question 2 of 4 [12 marks]

-11	-7	-4	-1	3	k	11	7	4	9	k	23
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Signal, S

-1	2	-1
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Kernel 1, k_1

$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
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Kernel 2, k_2

Use $k = \text{mod}(\text{Student ID}, 9) + 1$ for this question.

You have to work out only one operation for all the questions.

- (a) [3 marks] Find $S * k_1$ (Regular operation)
- (b) [3 marks] Find $S \star k_1$ (Regular operation)
- (c) [3 marks] Find $S * k_2$ (Truncated operation)
- (d) [3 marks] Find $S \star k_2$ (Truncated operation)

■ Question 3 of 4 [9 marks]

Channel 1

-11	-7	-4	-1	3	k	11	7	4	9	7	23
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Channel 2

-9	-3	-1	3	5	7	k	3	4	9	11	6
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Channel 3

7	9	15	17	21	13	11	k	4	-3	-13	-7
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The signal provided in this question has been collected from a 3 channel EEG headset and the scale is in μV . You need to extract the slow moving information using a low pass filter (moving average filter) with length of 3. Design the filter and apply it on the signal. In your solution, you should draw the filter and the output feature. Show one step of the calculation. Finally pass the output through a ReLU operation and show the final output.

For calculation, use truncated operation.

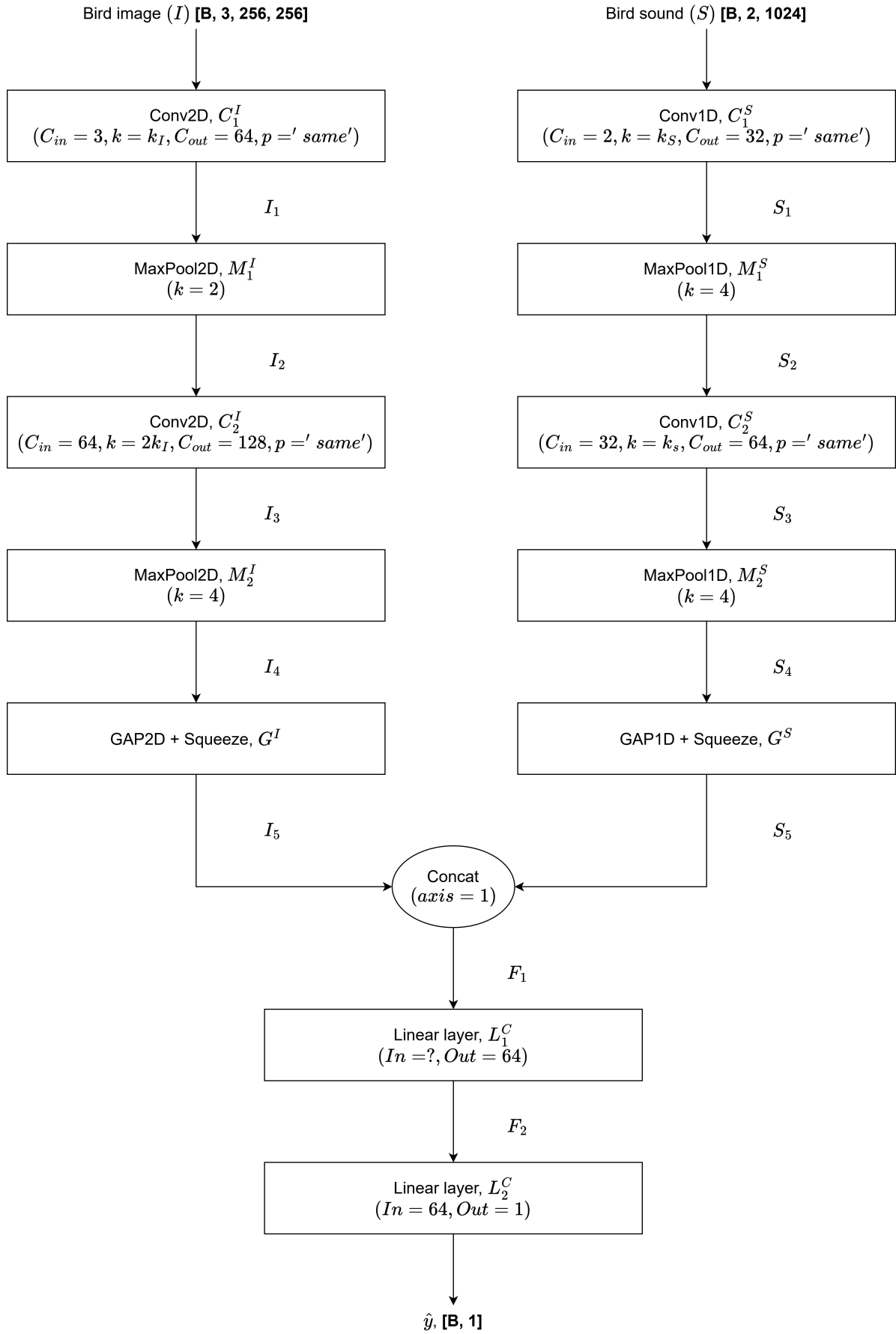
Also, for this question use $k = \text{mod}(\text{Student ID}, 7) + 3$

■ Question 4 of 4 [10 marks]

Ohi wanted to prepare a multimodal model to match bird calls with bird images. To this end, he prepared a convolutional neural network as provided below. He wants to figure out the parameter counts per layer and the dimensions of the intermediate outputs. Help him in figuring out these values by filling up the table provided at the end.

For this question use $k_I = 2 \cdot \text{mod}(\text{Student ID}, 3) + 3$ and $k_S = 2 \cdot \text{mod}(\text{Student ID}, 5) + 3$ and batch size, $B = 64$.

Assume bias = True for all relevant layers.



	Layer	Parameter count	Output variable	Dimensions
Image line	C_1^I		I_1	
	M_1^I		I_2	
	C_2^I		I_3	
	M_2^I		I_4	
	G^I		I_5	
Sound line	C_1^S		S_1	
	M_1^S		S_2	
	C_2^S		S_3	
	M_2^S		S_4	
	G^S		S_5	
Combined line	L_1^C		F_1	
	L_2^C		F_2	